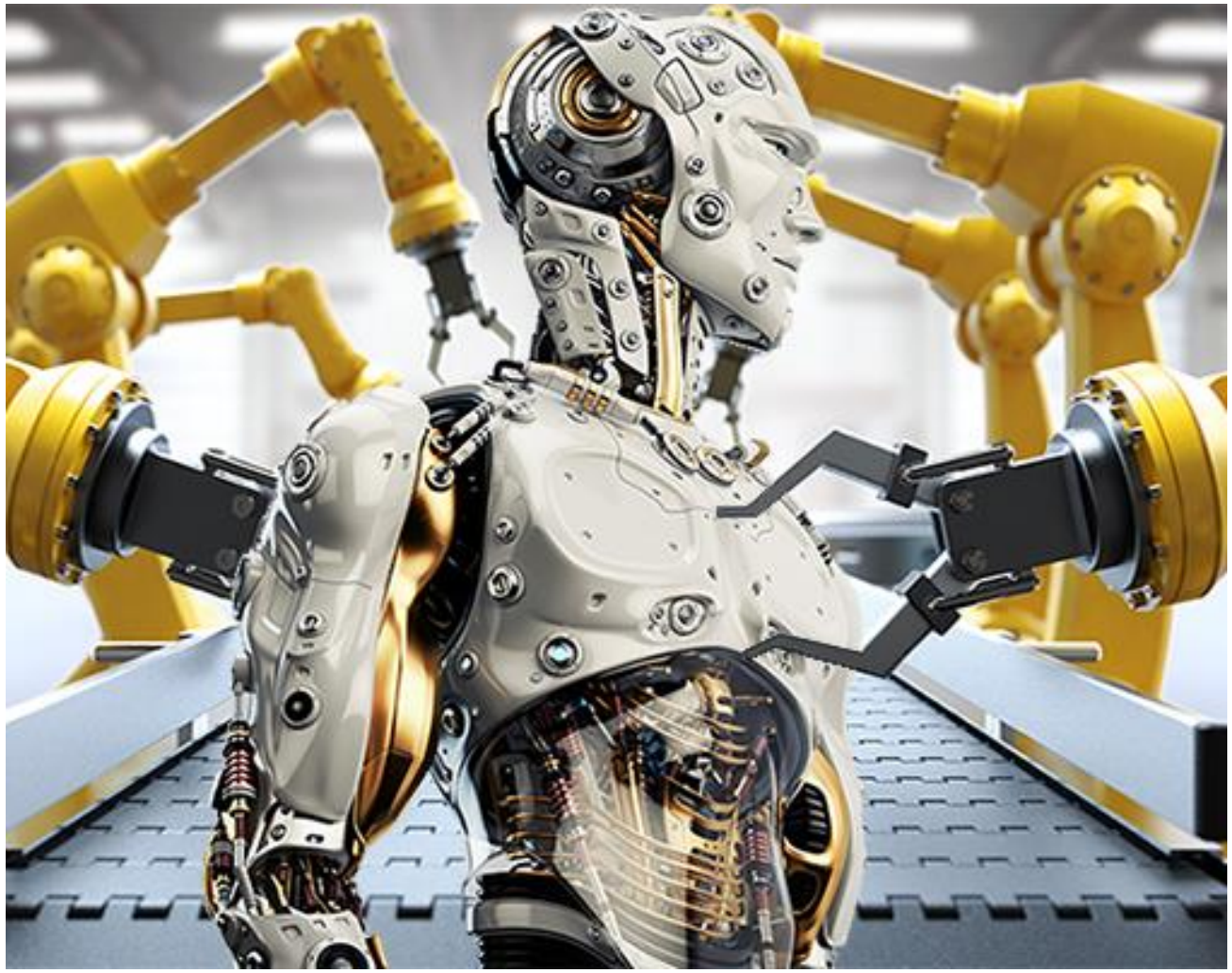
Welcome to

ROBO 1010
Introduction to Robotics and Automation

Sonia Chacko, PhD.



Lecture 1: Introduction

Outline

1. Introduction
2. Course Overview
3. Automation and Robotics – An Introduction

Course Plan

Week of	Lecture (Mon 11:00 AM - 12:20 PM)	Lab (We 2:30PM - 4:20PM)
8/25/2025	Course and Syllabus Introduction-	Basic Introduction to Automation Studio
9/1/2025	Labor Day – No Classes	Getting Familiar with the Automation Studio Exercises
9/8/2025	Electrical and Electronics Fundamentals	Electrical Circuit Fundamentals
9/15/2025	Basics of Electrical Sensors	Electric Circuits Using Sensors
9/22/2025	Basics of Electric Actuators	Circuits Using Electrical Actuators
9/29/2025	**Mid-term Exam**	*Relay and Logic Circuits Lecture*
10/6/2025	Fall Break – No Classes	Relay and Logic Circuits Lab
10/13/2025	Hydraulic Circuits – key concepts	Introduction to Hydraulic Circuit
10/20/2025	Pneumatic circuits – key concepts	Introduction to Pneumatic Circuit
10/27/2025	Pneumatic circuits – key concepts	Electro-Pneumatics Lab
11/3/2025	Intro to Control and Automation Systems	Basic Control System Lab
11/10/2025	Veterans Day – No Classes	Basic Control System Lab
11/17/2025	Control and Automation Systems	Automation Case Study Work
11/24/2025	Introduction to Robotic Systems	Nov. 26-28: Thanksgiving Break – No Classes
12/1/2025	Robot Programming Basics	Robot Programming Lab
12/8/2025	**Final Examination Week**	

Course Objectives

- **LO1:** Identify and describe key industrial components and systems used in robotics and automation.
- **LO2:** Explain and analyze the operation of fundamental mechanical and electronic components within automated systems.
- **LO3:** Integrate sensors and actuators to develop simple automated systems while considering safety, ethical use, and potential societal impacts.
- **LO4:** Apply foundational programming skills to control and automate system operations.
- **LO5:** Design and troubleshoot basic automation systems by conducting tests, analyzing results, and applying engineering judgment.

Key aspects of this course

Building Blocks of Electronics

- Learn the basics: voltage, current, resistance
- Discover how electricity powers automation
- Connect theory to real-world circuits

Sensors and Actuators

- Explore basics of sensors
- See how actuators (motors, pneumatics, hydraulics) make machines move

Control and Automation

- Introduction to control systems and PLCs
- Combine electronics, sensors, and actuators into working systems

Robotics Fundamentals

- Understand how robots fit into automation
- Get started with robot programming

What is Automation?

Automation:

The use of technology to perform tasks or processes with **minimal human intervention** to improve efficiency, accuracy, and safety.

Examples:

- Conveyor belt systems, automated assembly lines, smart home automation
- Robotic process automation (RPA) for office/business workflows – UiPath, BluePrism etc.



Industrial Automation

- Use of control systems, robotics, and digital technologies to operate machinery and production processes with minimal human intervention.

Benefits

- Higher Productivity & Efficiency
- Consistent Quality
- Worker Safety



Automation Example- Automated Wall Painting



Painting by Painters



Automated Painting

Automation Example – Smart Home Automation

- Smart Home Automation: Integration of everyday household devices into a connected network that can be monitored and controlled automatically.
- E.g.: Lighting & Climate Control, Security & Surveillance (smart locks, cameras) Voice & App Integration (Alexa, Google Home, Siri)

Benefits

- Convenience & Comfort
- Improved Security
- Energy Efficiency
- Remote Access & Control



What is Robotics?

Robotics:

A branch of engineering and science focused on designing, building, and operating robots to perform tasks.

Examples: Robotic arms in manufacturing, medical robots, warehouse robots, humanoid robots..





Automation Vs Robotics?

Automation:

The use of technology, such as control systems and software, to perform tasks or processes with **minimal human intervention**, to improve efficiency, accuracy, and safety and productivity.

Robotics:

A branch of engineering and science focused on designing, building, and operating robots—**programmable machines** capable of sensing, decision-making, and performing physical tasks autonomously or semi-autonomously.

Automation ≠ Robotics (automation is about processes, robotics is one way to achieve automation).

Key Difference:

- **Automation** can exist without robots (e.g., traffic light control).
- **Robotics** often involves automation but centers on physical, programmable machines. (e.g., Robotic arms, humanoid robots, autonomous drones)

“So… is a chatbot a robot or not?”



Traditional Robot:

- Robot is a machine with **physical embodiment** that can sense, process, and act on the world (e.g., robotic arm, mobile robot).
- Robots combine **sensors + actuators + control** to perform physical tasks.

- A chatbot is a software-based agent (**“software robot”**) that interacts with humans using natural language (text or voice). It processes information and gives responses but does not act physically in the real world.
- **Not a physical robot** → because it doesn’t have a body, sensors, or actuators. Chatbots fall under *intelligent agents* or *virtual robots*.

Robots = Physical + Digital Intelligence
Chatbots = Digital Intelligence only

Core Elements of Automation and Robotics Covering in this Course

